Maillard Reaction in a Glucose- Glycine system

Purpose

To observe the color changes during the water-heating bath of glucose and glycine, to explain the chemical reactions different composition with control group.

Materials

Sample	Glucose	Glycine	0.1% (w/v) sodium carbonate solution	Final volume
A: Mixed reaction sample	1.60 g	0.64 g	2.00 mL	8.00 mL
B: Glucose control	1.60 g	None	2.00 mL	8.00 mL
C: Glycine control	None	0.64 g	2.00 mL	8.00 mL
D: Mixture without sodium carbonate	1.60 g	0.64 g	None	8.00 mL

Observation

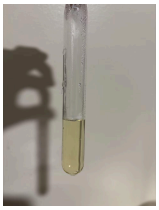
Sample A shows a deep brown color; it is the darkest solution among them. (figure1)

Sample B shows a light yellow color in the solution. Third darkest among the solutions. (figure2)

In sample C shows transparent and cololess mong the solutions.

In sample D, it is a dark brown color, after sample A.

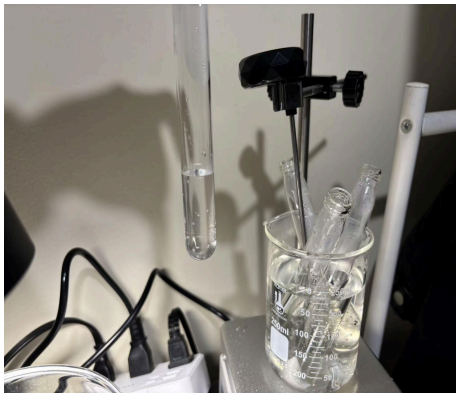

(figure1)


(figure2)


(figure3)


(figure4)





Chemical explanation

The Maillard reaction is a series of non-enzymatic reactions between glucose and the amino acid glycine. Heating facilitates the transformation of the intermediates into yellow and brown as the product of the reaction. Glucose is sometimes open-chain in water and has a CHO group at the end, which is reactive with the amino group of glycine and undergoes Amadori rearrangement; the reaction includes fragmentation and dehydration, causing the solution to absorb light and produce a dark color. The reason for the sample B is the color only result the degradation of sugar and sample C which lack of glucose cause the absence of the open chain tail CHO, where no reaction occurred. Sample D is that, even though there is no Sodium carbonate, the condition of the environment changed, causing less obvious color changes. Sodium carbonate makes the solution more alkaline: $CO_3^{2-} + H_2O \bullet HCO_3^- + OH^-$. This environment promotes sugar degradation nd easier for the reaction to occur.